Interactive Physics & Computational Simulations Suite

Theoretical Rigor, Symplectic Numerical Engines & Tactile Visual Laboratories in the Browser

11
Interactive Simulators

0
Build / NPM Dependencies

60 FPS
Canvas & WebGL Engines

Native
Web Audio Sonification

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Human-in-the-Loop, AI-Augmented Scientific Architecture

1. Theoretical Rigor
First-Principles Derivations

Every simulator begins with mathematical formulations: deriving PDEs, continuous Hamiltonian phase flows, Townsend electron avalanche kinetics, or 4D Riemannian metric tensors.

KEY AREAS:
Navier-Stokes * Poisson Statistics * Differential Geometry

2. AI Scaffolding & Velocity
Google Antigravity & Gemini

Leveraging modern AI agents for rapid architectural scaffolding, converting physical differential equations into web rendering primitives and procedural shader boilerplate.

KEY AREAS:
High Architectural Velocity * Zero-Dependency Single Files

3. Iterative Tuning & Physics
Symplectic & Numerical Verification

Hand-tuning integrators: migrating from naive Euler to symplectic Velocity Verlet and adaptive RK4, auditing energy drift, debugging boundary conditions, and designing tactile glassmorphism.

KEY AREAS:
Symplectic Integrators * Audio Sonification * Video Breakdowns

The Physics & Computational Spectrum: 6 Core Pillars

01. GEOMETRY & RELATIVITY
4D Manifolds & Relativistic Dynamics

Non-Euclidean SO(4) hyper-rotations, stereographic projection, Hopf Clifford Torus, and Schwarzschild event horizon light bending.

02. ARTIFICIAL LIFE & SWARMS
Aetheria: Terrarium (2D) & Cosmos (3D)

Quadtree-accelerated Craig Reynolds boids, neural predators, multi-segment IK sinusoidal leviathans, and N-body gravitational fields.

03. GEOPHYSICS & HYDRODYNAMICS
Glacial Lake Outburst Flood (GLOF)

Saint-Venant 1D/2D shallow water hydrodynamic wave equations, moraine dam breach rate modeling, and Himalayan valley impact.

04. NUCLEAR METROLOGY
Geiger-Muller Virtual Physics Lab

Townsend avalanche ionization cascades, Geiger plateau, dead time paralysis corrections, and Poisson-to-Gaussian counting statistics.

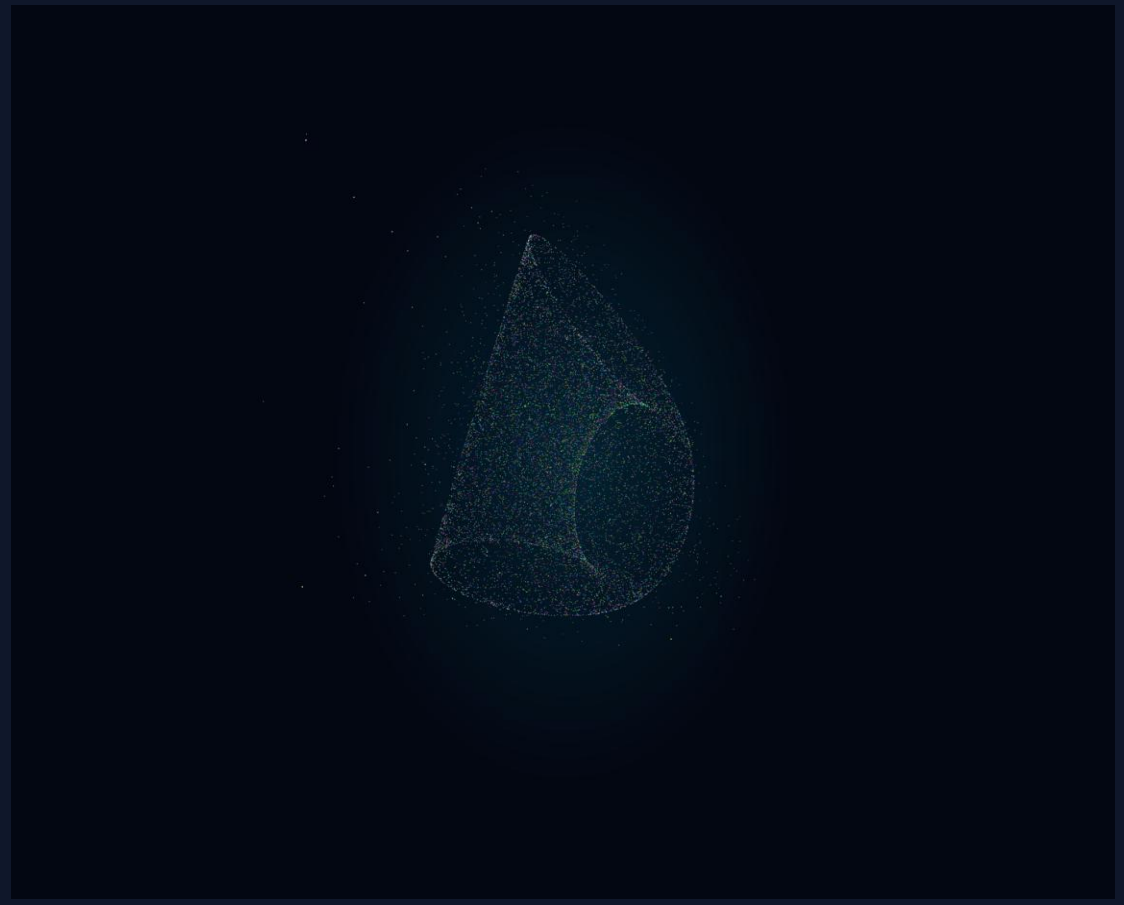
05. NONLINEAR DYNAMICS & CHAOS
Chaos as Music & Phase Space Lab

Lorenz & Rossler strange attractors, Lyapunov exponent tracking, Liouville volume invariance, and coordinate-to-pitch sonification.

06. NUMERICAL ANALYSIS
ODE Integrators & Newton-Raphson

Symplectic Velocity Verlet vs RK4 benchmarking, phase space orbit drift, and complex plane Newton fractal basin boundaries.

4D Manifold Projection & Relativistic Dynamics



Tech: HTML5 Canvas * Web Audio API * SO(4) Matrix Engine

MATHEMATICAL FORMULATION
Stereographic Projection R^4 -> R^3:
(x, y, z) = (X, Y, Z) / (1 - W/R)
SO(4) Non-Euclidean Hyper-Rotations across XW, YW, ZW

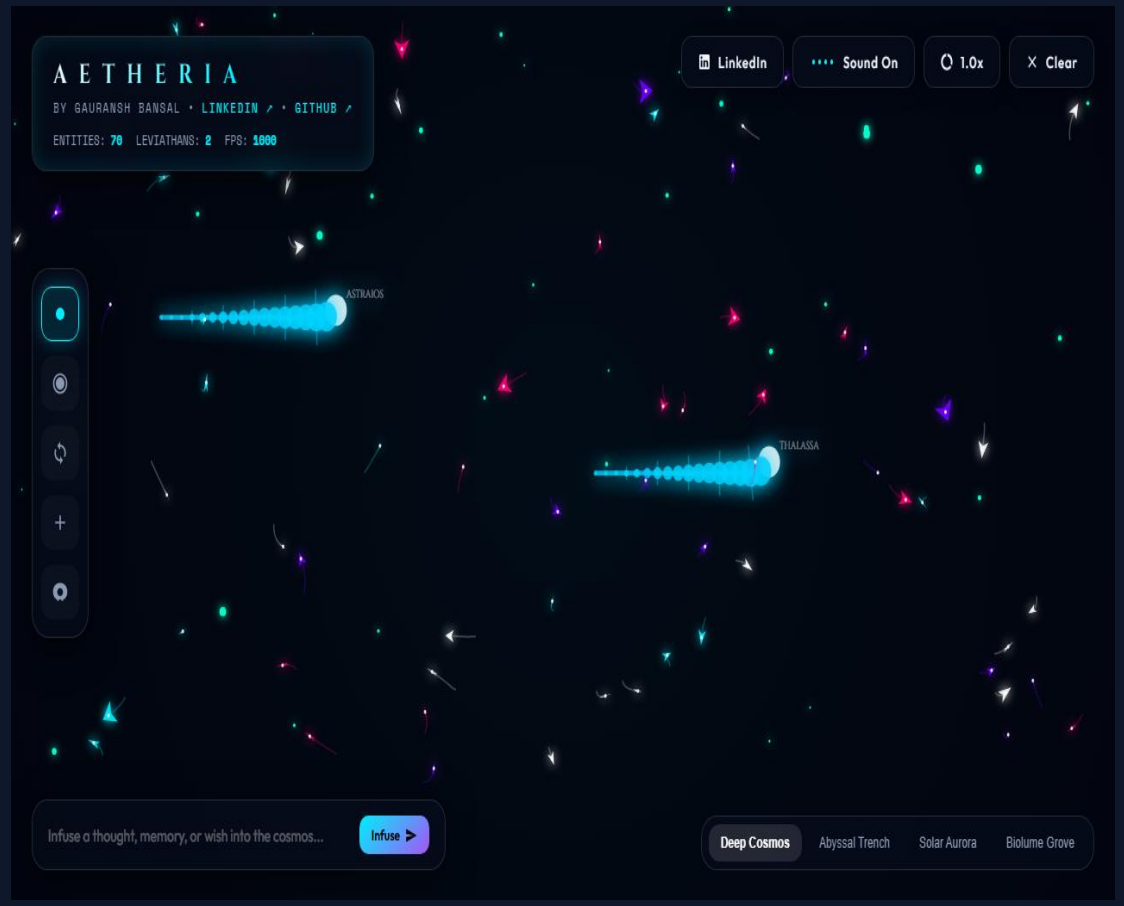
[*] Hopf Fibration & Tesseracts: Real-time projection of 4D hypercubes and Clifford Tori (S^3 -> S^2) into interactive 3D coordinates.

[*] Schwarzschild Gravitational Horizon: Dynamic event horizon exhibiting relativistic gravitational lensing and Doppler redshift/blueshift.

[*] 12,000 Relativistic Quantum Nodes: High-performance interactive particle point-cloud running at continuous 60 FPS.

[*] Polyphonic Curvature Sonification: Native Web Audio synthesizer generating harmonic resonances tied dynamically to local metric curvature.

AETHERIA: Terrarium (2D Generative Artificial Life)



Tech: HTML5 2D Canvas * Recursive Quadtree * Web Audio Synthesizer

BOID FLOCKING MECHANICS & KINEMATICS
F_total = w_s * F_sep + w_a * F_align + w_c * F_cohesion
Quadtree Spatial Search: O(N log N) vs Naive O(N^2)

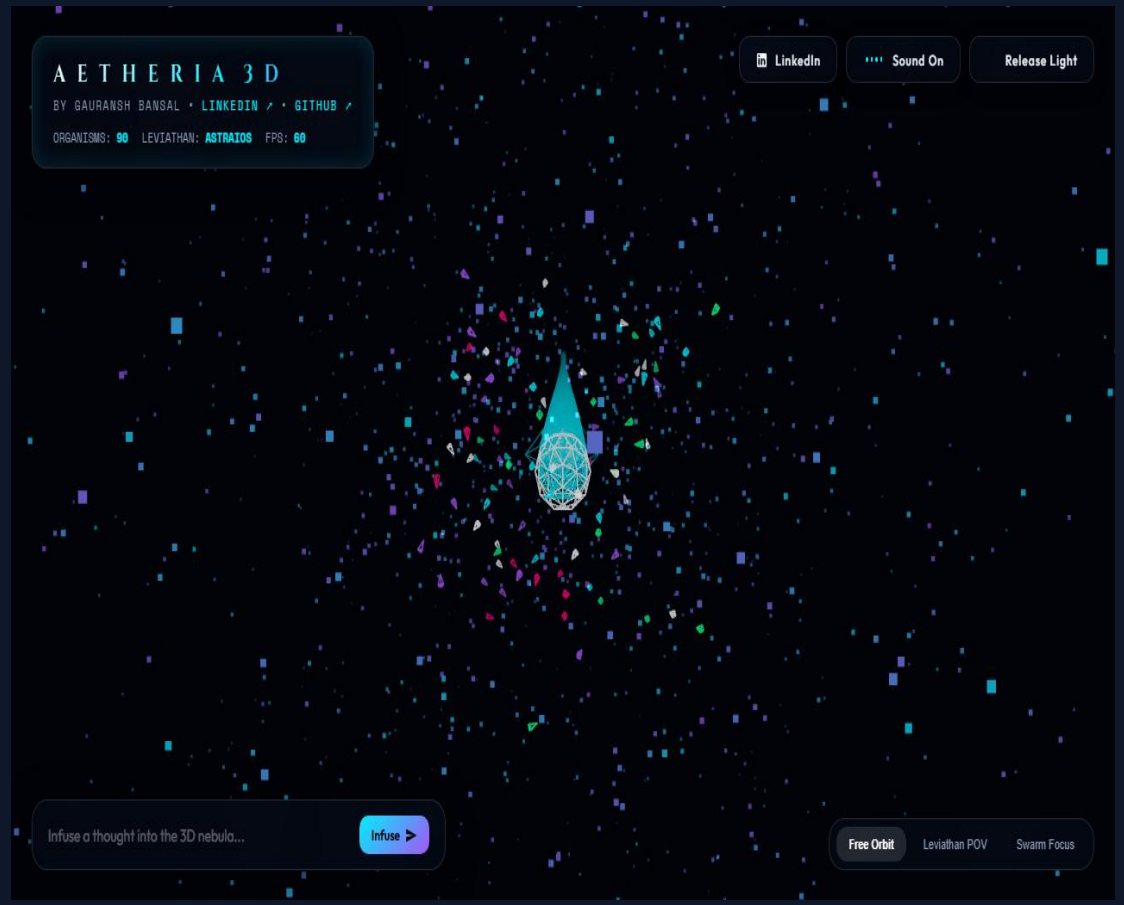
[*] Emergent Collective Intelligence: Autonomous agents exhibiting realistic cohesion, velocity alignment, and obstacle avoidance.

[*] Sinusoidal Inverse Kinematics: Multi-segment apex predator leviathans executing smooth predatory hunting trajectories.

[*] Evolutionary Genetic Drift: Speed, perception radius, and metabolic rates mutate and inherit across generations based on foraging success.

[*] Generative Pentatonic Audio: Procedural ambient synthesizer triggering harmonic tones synchronized to agent reproduction and predation events.

AETHERIA: Cosmos (3D Generative Ecosystem)



Tech: WebGL / Three.js * Post-Processing Bloom * Spatial Audio

N-BODY GRAVITATION & PLUMMER SOFTENING
a_i = SUM_j [G * M_j * (r_j - r_i) / (|r_j - r_i|^2 + eps^2)^(3/2)]
Quaternion Orientation & 3D Obstacle Avoidance

[*] Three.js WebGL Architecture: Full 3-dimensional multi-tiered gravitational potential wells with volumetric cosmic stardust nebulas.

[*] 3D Swarm Dynamics: Autonomous agent pods steering in 6 degrees of freedom with quaternion orientation vectors avoiding celestial bodies.

[*] Articulated Cosmic Leviathans: Segmented spine creatures with illuminated luminous energy blooms navigating deep cosmic space.

[*] Positional 3D Audio: Binaural spatial panning relative to interactive orbit controls and camera distance.

Glacial Lake Outburst Flood (GLOF) Simulator



Tech: HTML5 2D Canvas * Numerical PDE Integrator * Dynamic Graphing

SAINT-VENANT SHALLOW WATER WAVE EQUATIONS

$$\frac{\partial h}{\partial t} + \frac{\partial (uh)}{\partial x} = 0$$

$$\frac{\partial (uh)}{\partial t} + \frac{\partial}{\partial x}\left[u^2 h + \frac{1}{2} g h^2\right] = -g h \frac{\partial z_b}{\partial x} - \frac{\tau_b}{\rho}$$

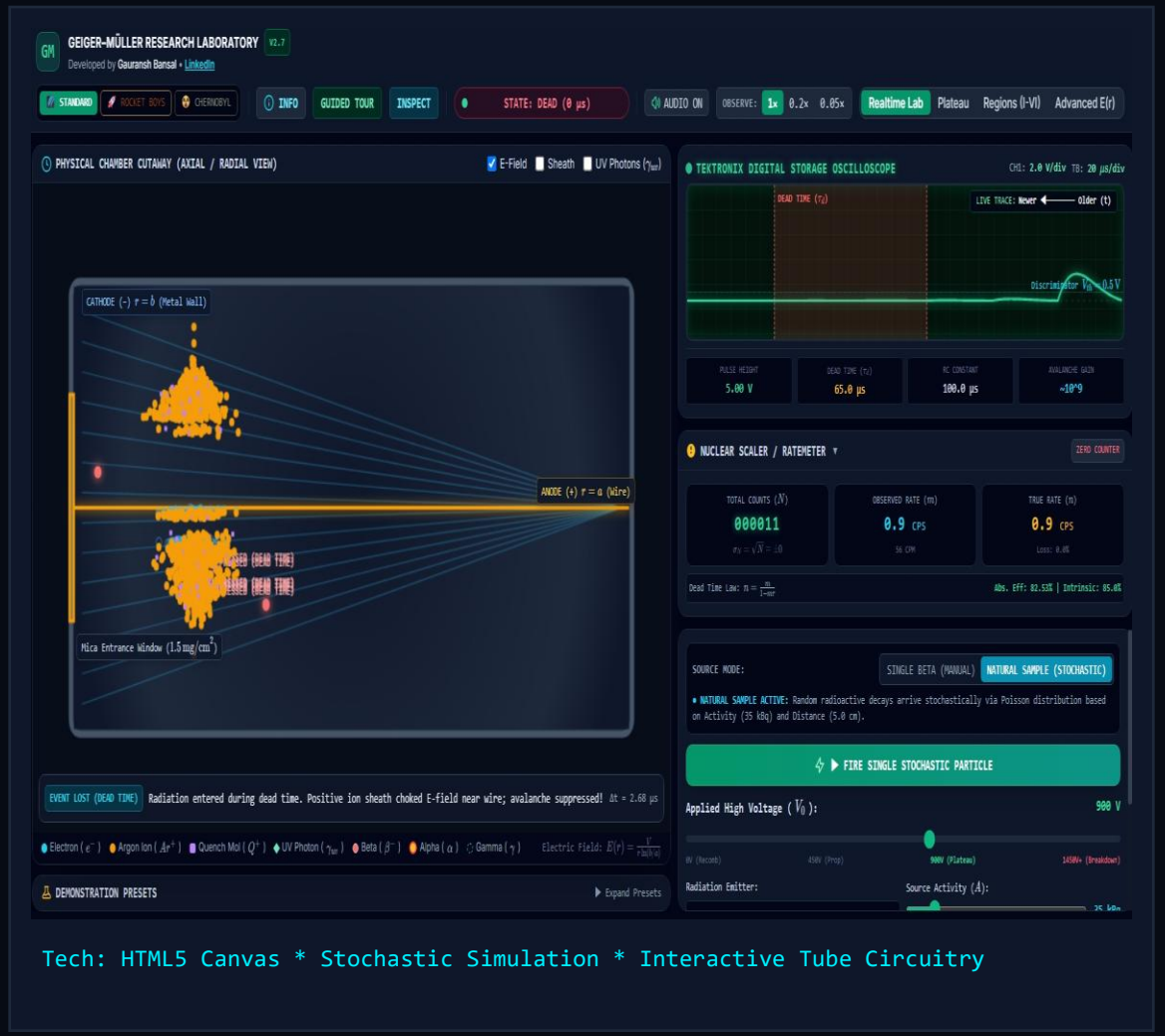
[*] Moraine Dam Breach Erosion: Dynamic breach widening and hydrograph discharge modeling calibrated with Himalayan breach records.

[*] Trishuli River Valley Topography: Digital elevation profile modeling downstream flood surge transit velocities and attenuation.

[*] Transient Discharge Graphing: Real-time calculation of peak discharge Q_peak, stage heights, and downstream inundation hazard zones.

[*] Video Breakdown Featured: Scripted scientific analysis published on LinkedIn examining climate risk in high-altitude glaciated basins.

Geiger-Muller Virtual Physics Laboratory



Tech: HTML5 Canvas * Stochastic Simulation * Interactive Tube Circuitry

TOWNSEND AVALANCHE & DEAD TIME RECOVERY
Ionization: n(x) = n_0 * exp(alpha * x)
Dead Time Correction: N_true = N_obs / (1 - N_obs * tau)
Poisson Distribution: P(k) = (lambda^k * e^(-lambda)) / k!

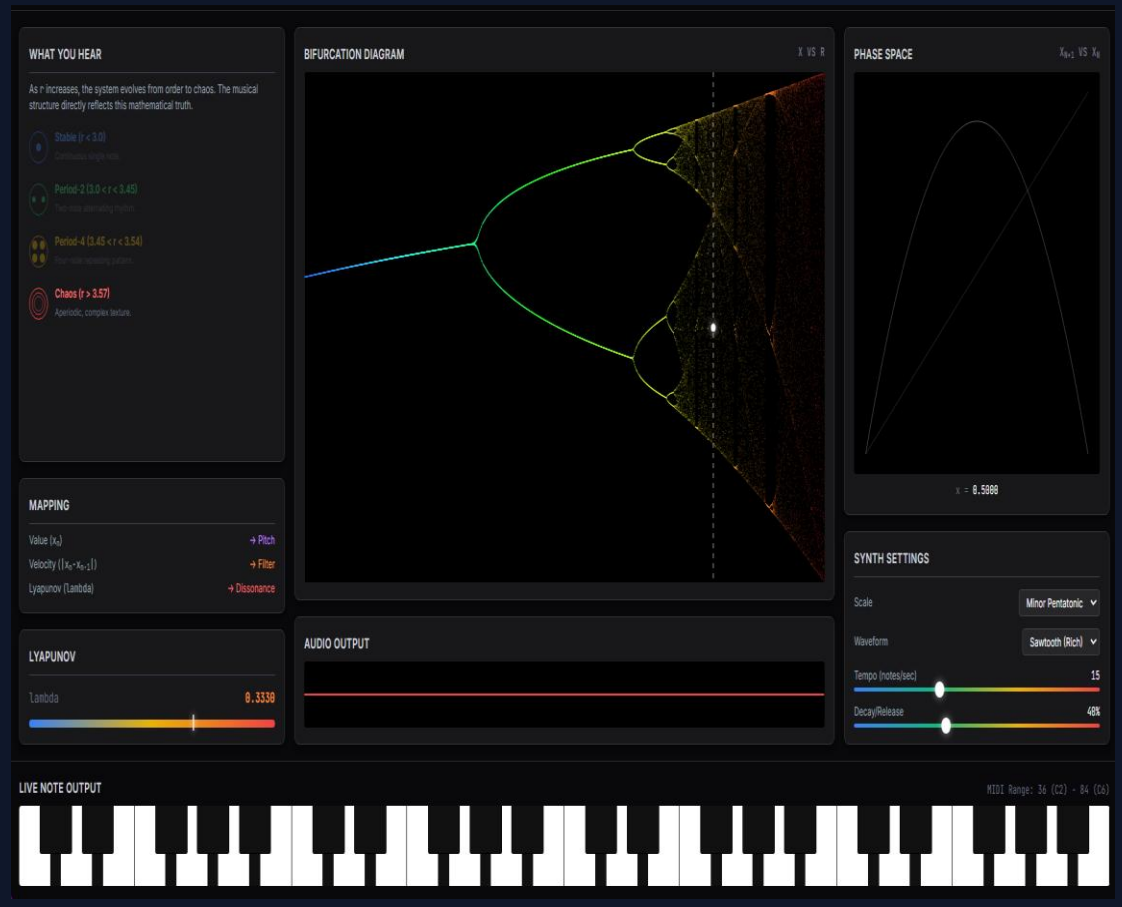
[*] Full Characteristic I-V Curve: Starting potential, threshold, Geiger plateau, and continuous Townsend avalanche discharge.

[*] Dead Time Kinetics: Simulates tube paralysis and non-paralysable counting corrections under intense radioactive fluxes.

[*] Statistical Convergence: Demonstrates Poisson count distribution approaching a Gaussian normal curve as mean count rate increases.

[*] Inverse-Square Radiation Law: Accurate distance attenuation measurements for selectable Alpha, Beta, and Gamma isotope sources.

Chaos as Music & Nonlinear Attractors



Tech: 3D Projection Engine * Web Audio Polyphonic Synth * Lyapunov Metric

LORENZ SYSTEM & SENSITIVE DEPENDENCE
dx/dt = sigma*(y - x), dy/dt = x*(rho - z) - y, dz/dt = x*y - beta*z
Divergence: ||delta x(t)|| ~= ||delta x(0)|| * exp(lambda * t)

[*] Strange Attractors in 3D: High-precision trajectory tracing of both classic Lorenz butterfly and Rossler spiral band attractors.

[*] Algorithmic Audio Sonification: Direct mapping of coordinate values (x -> pitch, y -> filter cutoff resonance, z -> stereo panning).

[*] Lyapunov Exponent Tracking: Real-time numerical estimation of positive maximal Lyapunov exponent proving deterministic chaos.

[*] The Butterfly Effect: Dual-particle injection with infinitesimal 10^-6 offset demonstrating rapid exponential trajectory divergence.

Phase Space Laboratory



Tech: HTML5 Vector Field Canvas * Symplectic Flow Engine * Topology Map

HAMILTONIAN FLOW & LIOUVILLE INVARIANCE
dq/dt = dH/dp, dp/dt = -dH/dq
Liouville Phase Volume Conservation: div(v_phase) = 0
Van der Pol Oscillator: d^2x/dt^2 - mu*(1 - x^2)*dx/dt + x = 0

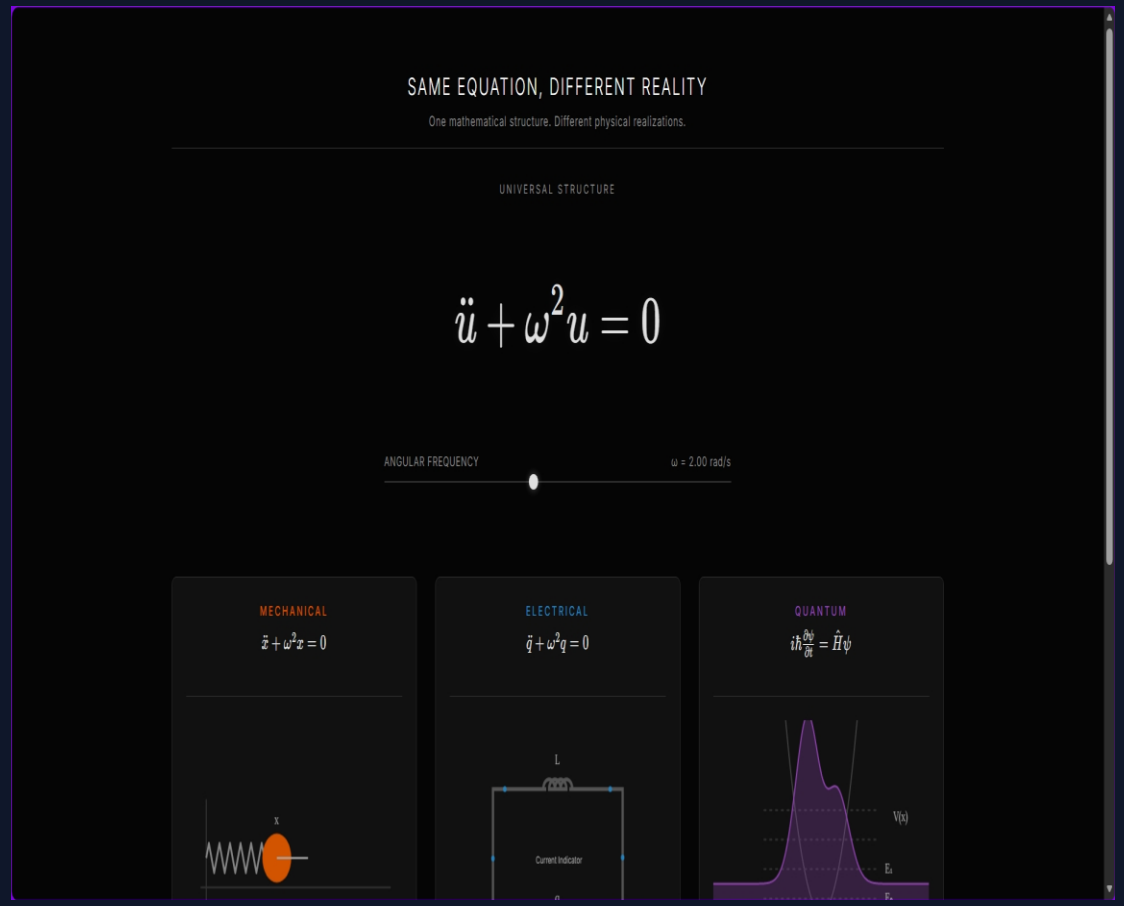
[*] Interactive Vector Fields: Visualizes velocity vector directions across generalized coordinate-momentum (q, p) phase space.

[*] Topological Fixed Points: Instant visual classification of hyperbolic saddle points, stable/unstable spirals, and elliptic centers.

[*] Nonlinear Limit Cycles: Emergence of isolated closed phase trajectories in dissipative self-excited systems like Van der Pol.

[*] Multi-Particle Ensemble Flow: Injects swarms of initial states to demonstrate conservation of phase space volume under Liouville theorem.

Same Equation, Different Reality



Tech: Multi-System Canvas Engine * Real-Time Resonance Curves * Audio Sync

THE UNIVERSAL HARMONIC OSCILLATOR ODE
d^2(psi)/dt^2 + 2*gamma * d(psi)/dt + omega_0^2 * psi = F(t)
Quality Factor: Q = omega_0 / (2*gamma)

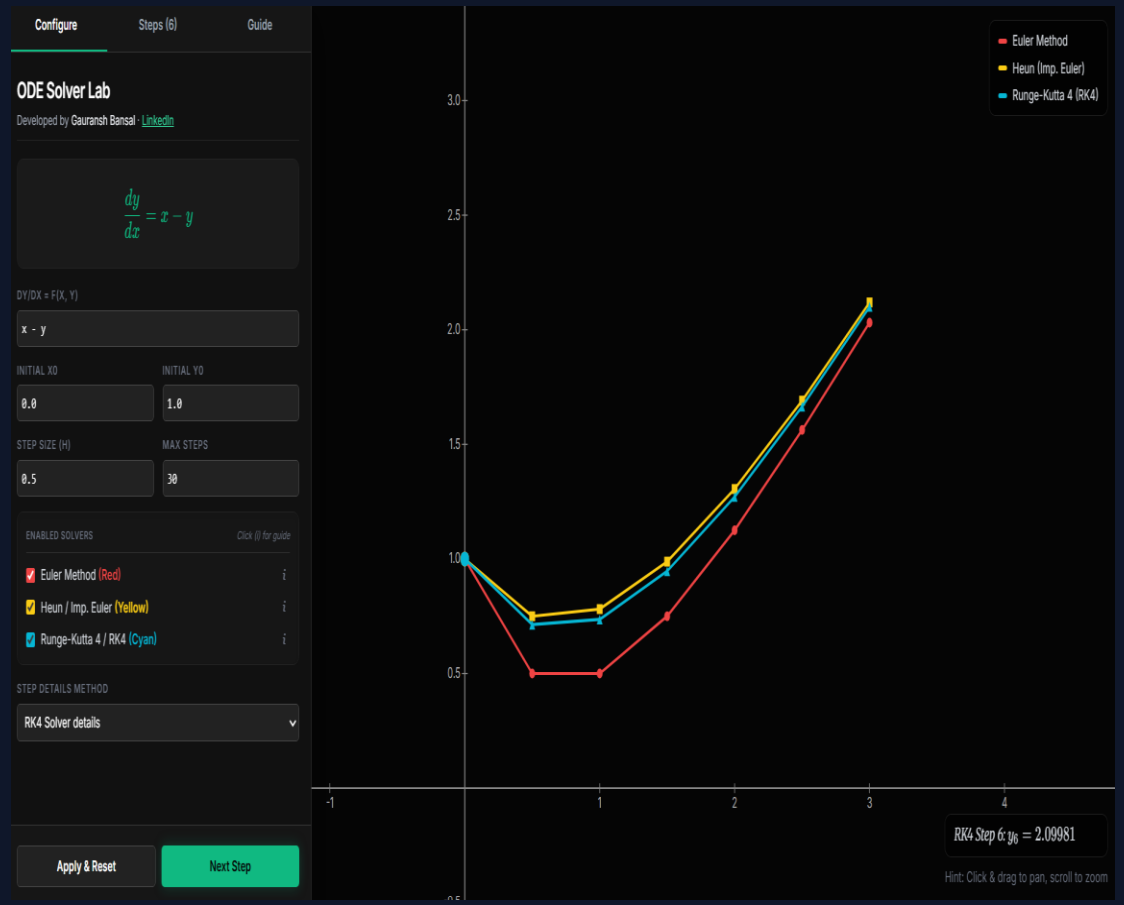
[*] Mechanical Domain: Damped driven pendulum / mass-spring: m*d^2x/dt^2 + b*dx/dt + k*x = F_0*cos(omega*t).

[*] Electrical Domain: Series RLC circuit: L*d^2q/dt^2 + R*dq/dt + (1/C)*q = V_0*cos(omega*t) (L <-> m, R <-> b, 1/C <-> k).

[*] Optical Domain: Lorentz electron oscillator modeling frequency-dependent dielectric permittivity and anomalous dispersion.

[*] Acoustic Domain: Helmholtz air cavity acoustic resonance and resonance curve amplification at fundamental frequencies.

ODE Numerical Integration Benchmark Lab



Tech: Benchmarking Harness * Phase Portrait Plotting * Real-Time Error Log

INTEGRATOR COMPARISON & ACCURACY ORDERS
Forward Euler: y_(n+1) = y_n + h * f(t_n, y_n) [O(h) - Unstable]
Euler-Cromer: v_(n+1) = v_n + h*a(x_n), x_(n+1) = x_n + h*v_(n+1) [Symplectic]
Runge-Kutta 4: [O(h^4) Precision] * Velocity Verlet: [Symplectic Shadow]

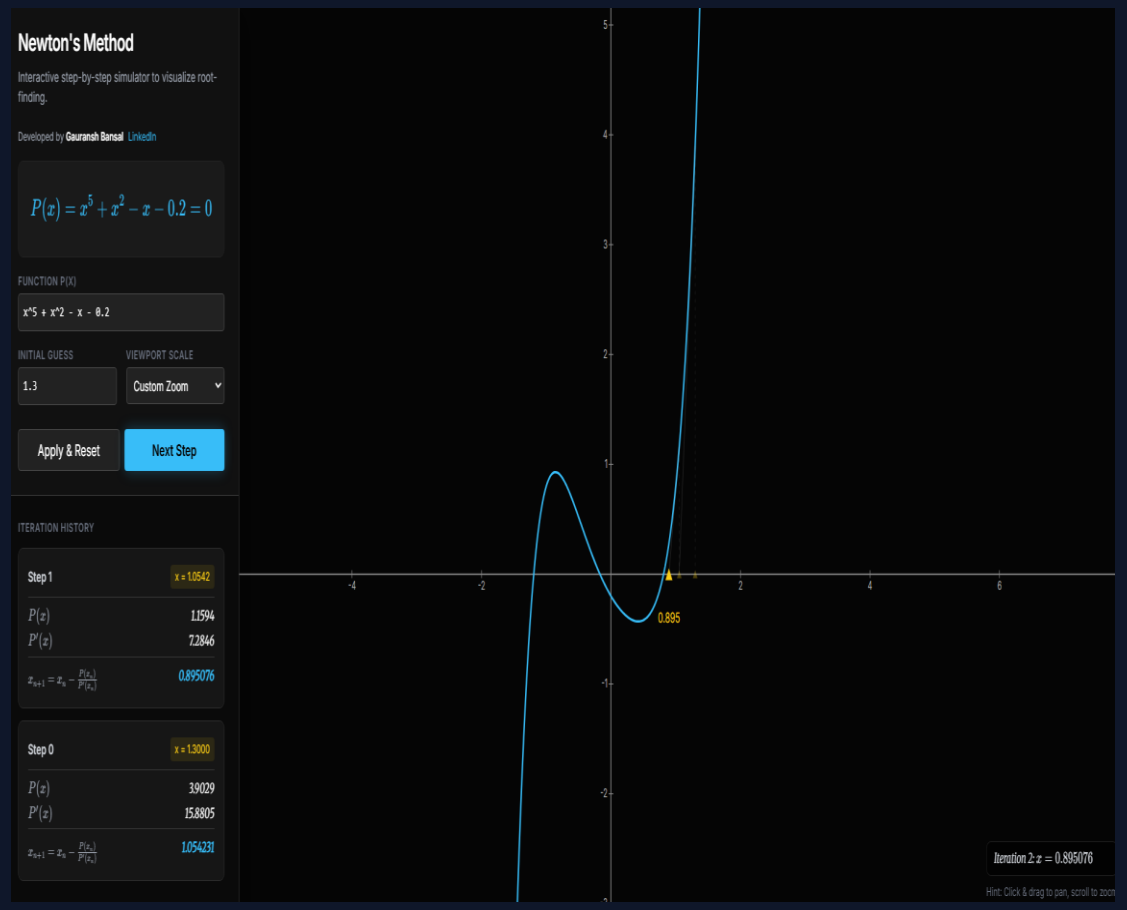
[*] Live Side-by-Side Benchmark: Subjects identical orbital systems to 4 foundational numerical solvers in real-time execution.

[*] Artificial Energy Explosion: Visually exposes how Forward Euler creates artificial outward energy spirals that violate conservation.

[*] Symplectic Invariance: Proves how Euler-Cromer and Velocity Verlet preserve phase space area and shadow Hamiltonians indefinitely.

[*] Secular Drift vs High Order: Demonstrates that RK4 offers supreme short-term precision but experiences secular energy drift over 10^6 steps.

Newton-Raphson & Root-Finding Dynamics



Tech: Interactive Function Parser * Tangent Animator * Complex Fractal Canvas

ITERATION FORMULA & QUADRATIC CONVERGENCE
x_(n+1) = x_n - f(x_n) / f'(x_n)
Quadratic Error Scaling: |e_(n+1)| ~= [|f''(r)| / (2*|f'(r)|)] * |e_n|^2

[*] Dynamic Tangent Line Geometry: Step-by-step interactive visual playback of tangent projections intersecting the x-axis.

[*] Quadratic Convergence Rate: Doubles significant digits of accuracy with each consecutive iteration near simple roots.

[*] Pathological Divergence Modes: Live demonstration of runaway explosions when f(x) -> 0 and infinite 2-cycle oscillation traps.

[*] Newton Fractal Basins: Extension to the complex plane z in C generating fractal boundaries with infinite self-similar complexity.

Landscape of Futures (LOF) -- Nonlinear Dynamics Explorer

MULTIVARIABLE POTENTIAL SURFACES & TIPPING POINTS
dx/dt = -grad V(x, mu) + eta(t)
Parameter shifts mu in R^k trigger Saddle-Node, Pitchfork, and Hopf Bifurcations

KEY CONCEPTUAL FRAMEWORK:

[*] Multidimensional Potential Landscapes: Visualizes dynamic potential wells V(x, y) where valleys represent stable societal, ecological, or physical equilibrium states.

[*] Stochastic Tipping Perturbations: Noise terms eta(t) simulating perturbations that knock system states across critical separatrices into alternative regime basins.

[*] Scenario Exploration: Interactive sliders warping the underlying energy topology to model cascading nonlinear transitions.

SYSTEMIC FEATURES & METHODOLOGY:

[*] Real-Time Potential Morphing:
Sliders deform the multi-well energy landscape in real time, demonstrating critical slowing down as attractors flatten near bifurcation thresholds.

[*] Probability Density Histograms:
Computes long-term Monte Carlo phase trajectories to generate empirical probability distributions across competing basins.

[*] Interdisciplinary Modeling:
Extends nonlinear dynamical systems theory directly to complex macroeconomic, ecological, and civilizational futures.

[*] Single-File Client-Side Engine:
Fully self-contained within lof.html with responsive Canvas graphics and reactive state controls.

Full-Stack Client-Side Architecture & Performance

Zero Build Overhead

No npm, Vite, Webpack, or bundlers. Every simulation is an autonomous, portable single-file HTML5 application ready for local execution or static hosting.

60 FPS Physics Pipeline

Sub-stepping physics loops decoupled from requestAnimationFrame. High-DPI coordinate scaling, batch canvas stroke calls, and optimized WebGL buffers.

Web Audio Sonification

Direct procedural oscillator graphs, biquad filter sweeps, and gain envelopes translating mathematical coordinates into rich acoustic signals.

Security & IP Protection

Built-in anti-framing framebusting protections, drag-prevention layers, domain locking, and console intellectual property fingerprint notices.

Cross-Browser Compatibility: Chrome, Edge, Firefox, Safari, iOS & Android * 100% Client-Side Physics Evaluation

Democratizing Physics: Scripted Video Breakdown Series

End-to-End Breakdown

For each simulator, Gauransh scripts, records, and edits a dedicated video breakdown explaining the physical derivations, PDEs, boundary conditions, and numerical mechanics.

Active LinkedIn Reach

Published regularly on LinkedIn to engage academic students, university professors, and software engineers in discussions on computational physics and science communication.

Interactive Pedagogy

Replacing dry blackboard textbook formulas with tactile, dynamic visual laboratories where students can tweak sliders, test edge cases, and hear resonance directly.

WATCH THE VIDEO BREAKDOWN SERIES ON LINKEDIN:
https://www.linkedin.com/in/gauransh-bansal-a71223239/
Follow along for ongoing breakdowns of GLOF outburst hydrodynamics, Geiger-Muller dead time, and chaos sonification.

Summary, Research Roadmap & Live Exploration

UPCOMING SCIENTIFIC ADDITIONS:

[1] Quantum Wavepacket Dynamics (1D/2D):
Numerical time-dependent Schrodinger equation solver [i*hbar*dPsi/dt = H*Psi] showing barrier tunneling and dispersion.

[2] General Relativistic Ray Tracing:
Null geodesic path integration around rotating Kerr black holes with dynamic gravitational lensing and accretion disks.

[3] WebGPU Compute Shaders:
Massively parallel particle swarms scaling to 1,000,000+ agents via browser-native compute shaders.

[4] Lattice Boltzmann Fluid Dynamics:
2D/3D Navier-Stokes vortex shedding past obstacle geometries.

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[*] GitHub Repository:
https://github.com/gauranshbansal

[*] Interactive Hub:
Open index.html to explore the live gallery.

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